TIER 1 PROJECT

# Pixel Predators

## Plant Disease Classification

ITAI 1378 | Midterm Project Proposal


TEAM MEMBERS

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Upload a leaf photo. Get a disease label and confidence score.

# The Problem

!

## >$220B

annual global losses from plant pests and diseases

%

## Up to 40%

of crop production lost to pests and diseases each year

?

## Manual

scouting can be slow, costly, and dependent on expert access

**Who cares:** Farmers, agronomists, food security groups, and crop insurance teams.

**What it solves:** Fast preliminary screening from a leaf photo, helping growers make earlier treatment decisions.

Sources: FAO Plant Production and Protection facts and figures; USDA ARS notes that early detection and rapid response are key to mitigating plant diseases.

# Your Solution

A transfer-learning app that classifies a leaf image and returns the predicted plant condition with a confidence score.

**IMG**

**Input**

Leaf photo
JPG / PNG

→

**224**

**Preprocess**

Resize 224x224
Normalize

→

**AI**

**Model**

ResNet50
feature extractor

→

**✓**

**Output**

Disease or healthy
+ confidence

**Application layer:** Gradio upload page - user selects a leaf image - app shows top prediction, confidence, and a simple report.

Model: ResNet50 pretrained on ImageNet | Framework: PyTorch + torchvision | Tier: 1

# Technical Approach

AI
**CV Technique**
Image classification using transfer learning

M
**Model**
ResNet50, pretrained on ImageNet; freeze most backbone layers and fine-tune the final classifier

FW
**Framework**
PyTorch + torchvision for training; Hugging Face or Kaggle for dataset access; Gradio for demo UI

✓
**Why this fits**
Classification matches the goal: one image in, one crop-condition label out. ResNet50 is strong, simple, and realistic on free Colab.

Approach adjusted to Tier 1: pretrained image classification model + application logic, not a full custom detector or segmentation system.

# Data Plan

DATASET

## PlantVillage Dataset

*Public dataset via Kaggle / TensorFlow Datasets / original PlantVillage source*

- ~54K healthy and diseased leaf images
- 38 crop-condition classes across 14 crop species
- Labels include crop name + disease/healthy status
- Mostly controlled leaf photos, so real-world testing is important
- Fallback: balanced 10-class subset if Colab limits hit

PREPROCESSING + SPLIT

01 **Resize**
224 x 224 px for ResNet50 input

02 **Augment**
Training only: flip, rotation, color jitter

03 **Normalize**
ImageNet mean and standard deviation

04 **Split**
70% train / 15% validation / 15% test

Sources: Mohanty et al. (2016), Using Deep Learning for Image-Based Plant Disease Detection; TensorFlow Datasets plant_village catalog.

# Success Metrics

PRIMARY METRIC

**90%+**

Top-1 test accuracy

F1

**Macro F1**

>= 0.88 average across classes

⏱️

**Speed**

< 1 second per image

R

**Recall check**

Review low-performing classes

CM

**Tools**

Confusion matrix + error samples

Final report will use the held-out test set. Validation will be used only for model tuning and parameter choices.

# Milestone Plan

6-week team plan from project setup to final presentation and submission.

| | | |
|---|---|---|
| Week 1 | **Create Team** | Form the Pixel Predators team and confirm team members |
| Week 2 | **Blueprint** | Review requirements, start the proposal, GitHub repo, and README |
| Week 3 | **Choose Project + Dataset** | Select plant disease classification and confirm the PlantVillage dataset |
| Week 4 | **Refine Project + Assign Tasks** | Finalize scope, assign team roles/titles, and divide slide/project tasks |
| Week 5 | **Improve Project** | Improve the model plan, metrics, risks, resources, and demo workflow |
| Week 6 | **Present and Submit** | Finalize slides, repository, proposal PDF, presentation, and Canvas submission |

This plan uses six project weeks; exact calendar dates should match Canvas.

# Risks and Resources

**Top Risk 1**

**PlantVillage images may not match outdoor field photos.**

Plan B: use stronger augmentation, test a few real leaf photos, and clearly state the app is a preliminary screening tool.

**Top Risk 2**

**Full dataset may exceed free Colab time or memory.**

Plan B: train a balanced 10-class subset, lower batch size, save checkpoints to Drive, and use Kaggle Notebooks as backup.

## Resources Needed

- Compute: Google Colab free tier; Kaggle Notebooks backup
- Storage: Google Drive for model checkpoints
- Tools: PyTorch, torchvision, Hugging Face Datasets, Gradio
- Data: PlantVillage public dataset
- Estimated cost: $0

**Fully achievable within the term using free tools.**

Backup scope keeps the project feasible while still meeting the Tier 1 requirements and rubric goals.